

Companion LXIX: Spectral Density Functions of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

Spectral density functions (SDFs) of persistence diagrams provide a frequency- resolved characterization of topological fluctuations in cosmological fields. In weak-lensing analyses, SDFs quantify how the birth and death of connected components, loops, and void-like structures contribute across frequency modes. In the standard cosmological model, the SDF reflects the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the spectral distribution of topological features in a characteristic way. This Companion develops a continuous description of SDFs in RDCC, deriving how the relaxation scale influences frequency-resolved topology, spectral coherence, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field through spectral topology.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. While cross-correlation functions quantify filtration-coupled topology, spectral density functions provide a frequency- resolved view of topological fluctuations.

In the standard cosmological model, the SDF reflects the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the spectral distribution of topological features in a scale-dependent manner, producing a distinctive signature in SDF analyses.

2 Spectral Density Functions of Persistence Diagrams

Given a time- or filtration-ordered sequence of persistence diagrams $\{D(\nu_i)\}$, the spectral density function is defined as the Fourier transform of the Wasserstein-distance fluctuations:

$$S_D(\omega) = \left| \sum_i W_p(D(\nu_i), D^*) e^{-i\omega\nu_i} \right|^2. \quad (1)$$

This function measures the frequency content of topological fluctuations.

3 RDCC Modification of Persistence Diagrams

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (2)$$

This modifies the birth and death thresholds of topological features:

$$b_{\text{RDCC}} = b [1 - \chi_b(\ell/\ell_{\text{rel}})^{\alpha}], \quad (3)$$

$$d_{\text{RDCC}} = d [1 - \chi_d(\ell/\ell_{\text{rel}})^{\alpha}]. \quad (4)$$

Thus the RDCC spectral sequence becomes

$$W_p(D_{\text{RDCC}}(\nu), D_{\text{RDCC}}^*). \quad (5)$$

4 Spectral Density Functions in RDCC

The RDCC spectral density function is

$$S_{D,\text{RDCC}}(\omega) = S_D(\omega) \left[1 - \chi_S \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (6)$$

This modification reflects the relaxation-driven change in frequency-resolved topology.

5 Spectral Topological Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Spectral density functions therefore exhibit:

- enhanced low-frequency power (large-scale coherence),
- suppressed high-frequency power (small-scale suppression),
- a characteristic turnover near ℓ_{rel} ,
- modified spectral slopes across redshift,
- altered frequency alignment of persistence lifetimes.

These signatures provide a direct probe of the RDCC gravitational field through spectral topology.

6 Conclusion

Spectral density functions of persistence diagrams in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the frequency-resolved topology in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of spectral topology, nonlinear structure, and the interplay between light and gravity in RDCC.

References

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